Definition. Let f be a function defined on an open interval containing a. The *derivative of f at a*, denoted by $f'(a)$, is

$$f'(a) = \lim_{h\to 0} \frac{f(a+h)-f(a)}{h},$$

provided this limit exists. If $f'(a)$ exists, we say that f is *differentiable at a*.

If f is differentiable at all points in some open interval I, we say that f *is differentiable on I*. If I is a closed interval, we say that f *is differentiable on I* if it is differentiable at all interior points of I as well as left-differentiable and right-differentiable at the endpoints. If f is differentiable on an interval I, we can define a function f', called the *derivative of f*, which assigns to each point $x \in I$ the derivative of f at x; that is, $f' : x \mapsto f'(x)$.

If we write $y = f(x)$, other frequently used notations (called *Liebniz notation*) for $f'(a)$ are $\frac{dy}{dx}(a), \frac{df}{dx}(a), \frac{dy}{dx}\big|_a$, and $\frac{df}{dx}\big|_a$. Note that the notation $\frac{dy}{dx}$ does *not* imply that the derivative is a quotient of quantities "dy" and "dx" - these (at least as of now) do not have independent meaning outside of the expression $\frac{dy}{dx}$ any more than "$\to$" has independent meaning outside of the expression $\lim_{x\to a} f(x)$. The function $f' : x \mapsto f'(x)$ is often denoted $\frac{d}{dx} : x \mapsto \frac{dy}{dx}(x)$.

1. Let f be the constant function $f(x) = c$.

   (a) Find $f'(x)$.

   (b) Recalling that $f'(x)$ is the slope of the tangent line to the graph of f at $(x, f(x))$, does your answer from part (a) make sense? Explain.

2. Let f be the linear function $f(x) = x$.

   (a) Find $f'(x)$.

   (b) Recalling that $f'(x)$ is the slope of the tangent line to the graph of f at $(x, f(x))$, does your answer from part (a) make sense? Explain.

3. Let f be the linear function $f(x) = 2x + 3$.

   (a) Find $f'(x)$.

   (b) Recalling that $f'(x)$ is the slope of the tangent line to the graph of f at $(x, f(x))$, does your answer from part (a) make sense? Explain.

4. Let f be the linear function $f(x) = mx + b$.

   (a) Find $f'(x)$.

   (b) Recalling that $f'(x)$ is the slope of the tangent line to the graph of f at $(x, f(x))$, does your answer from part (a) make sense? Explain.

5. Let $f(x) = -16x^2 + 96x$. Find $f'(x)$.

6. (a) Assume f is differentiable at a. Show that the equation of the tangent line to the graph of f at $(a, f(a))$ is given by $y = f'(a)(x-a)+f(a)$.

   (b) Let $f(x) = -16x^2+96x$. Find the equation of the tangent line to the point $(1,80)$. (Use your answer to question 5 to find $f'(1)$.)

7. Show that $f'(a)$ can also be written as $f'(a) = \lim_{h\to 0} \frac{f(x)-f(a)}{x-a}$. (*Hint*: Substitute $h = x-a$ and note that $h \to 0$ if and only if $x \to a$.)

Homework

Compute the equation of the tangent line to the graph of $f(x) = 3x^2 - 2x$ at the point $(-2, 16)$.